Checkpoint 6 Science — Practice Test 1

Paper 1: 35 minutes · calculator allowed · Paper 2: 35 minutes · Total 80 marks

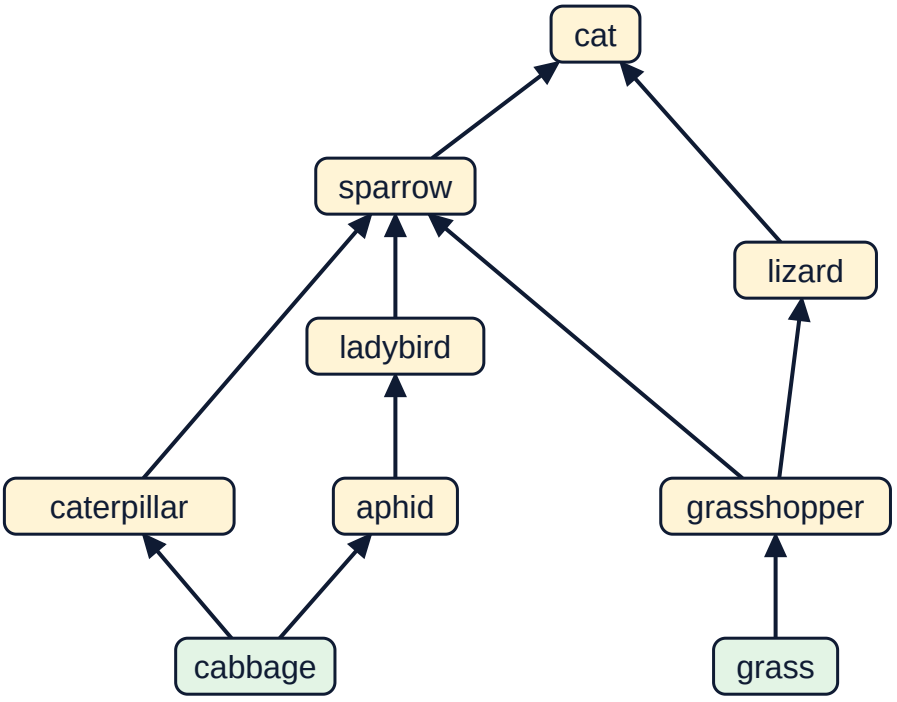
Answer all questions. Write your answers in the spaces provided. Show your working where there is space for it. The number of marks for each question is shown in brackets [ ].

Name _____ Class _____ Date _____ Mark: ____ / 80

Abdelkriem Academy · practice paper written in the style of Cambridge Checkpoint. Not an official Cambridge paper.

Paper 1 — 35 minutes — 40 marks

1 Pupils find these living things in the school garden. The food web shows what eats what. **[1]**



(a) What is the source of energy for this food web?

(b) How many food chains in this food web include the sparrow? **[1]**

Working

Answer

(c) Which food chain is in this food web? **[1]**

- ☐ A) cabbage → grasshopper → lizard → cat
- ☐ B) grass → grasshopper → lizard → cat
- ☐ C) aphid → cabbage → ladybird → sparrow
- ☐ D) grass → caterpillar → sparrow → cat

(d) The gardener sprays the cabbages with a pesticide to kill the aphids. The pesticide is toxic and it stays in animals' bodies. **[2]**
Explain why the cat could be harmed, even though it never eats cabbage.

2 The garden club wants to know if adding compost to sandy soil helps bean plants grow taller. They fill pots with sandy soil and mix in different amounts of compost. They plant one bean seed in each pot and use three pots for each amount of compost. After four weeks they measure the height of each plant. **[1]**

Compost added (spoonfuls)	Pot 1 height (cm)	Pot 2 height (cm)	Pot 3 height (cm)
0	12	13	11
2	17	18	16
4	21	9	22
6	24	25	23

(a) What is the independent variable in this investigation? (the variable the club changes)

- ☐ A) the height of the plants
- ☐ B) the type of soil
- ☐ C) the amount of compost
- ☐ D) the number of pots

(b) Give **one** variable the club must keep the same to make this a fair test. **[1]**

(c) One result is anomalous (it does not fit the pattern). Write down this height. **[1]**

Working

Answer ____ cm

(d) Calculate the mean height of the plants with 2 spoonfuls of compost. **[1]**

Working

Answer ____ cm

(e) Write a conclusion for this investigation. **[1]**

(f) Use your science to explain why adding compost helps plants to grow in sandy soil. **[1]**

3 Pupils take soil from three parts of the school garden and find out what each soil contains. **[1]**

Soil	Sand (%)	Clay (%)	Organic matter (%)
X	80	10	10
Y	15	75	10
Z	40	25	35

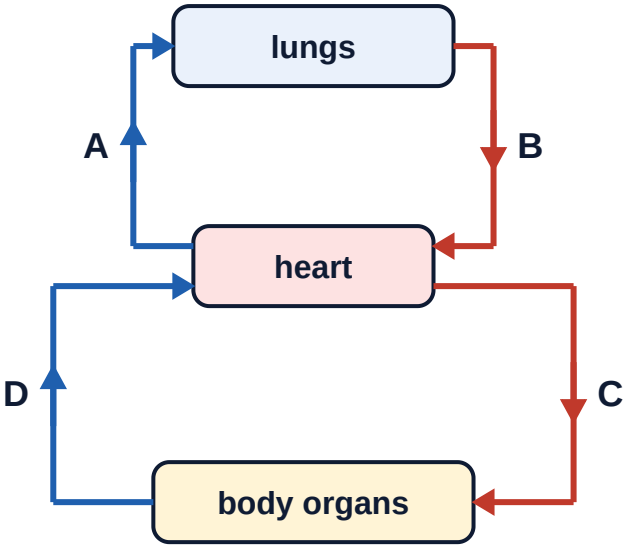
(a) Which soil is a clay soil? Write X, Y or Z.

(b) Water drains through one soil the fastest. Which soil? Write X, Y or Z. **[1]**

(c) Soil X dries out very quickly in summer. Suggest what the gardener could add to soil X to help it hold more water. **[1]**

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4 The diagram shows how blood flows around the human body. The arrows show the direction of blood flow. **[1]**



(a) What type of blood vessel is D?

- ☐ A) artery
- ☐ B) capillary
- ☐ C) trachea
- ☐ D) vein

(b) Which vessel, A or B, carries blood that has just collected oxygen? **[1]**

(c) Name the tiny blood vessels inside the body organs, where oxygen and nutrients pass from the blood into the cells. **[1]**

(d) Name the waste gas that the blood carries from the body cells to the lungs. **[1]**

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5 The school garden has a barrel of rainwater. In summer, mosquitoes lay their eggs in the still water. Some kinds of mosquito can carry the parasite that causes malaria. **[2]**

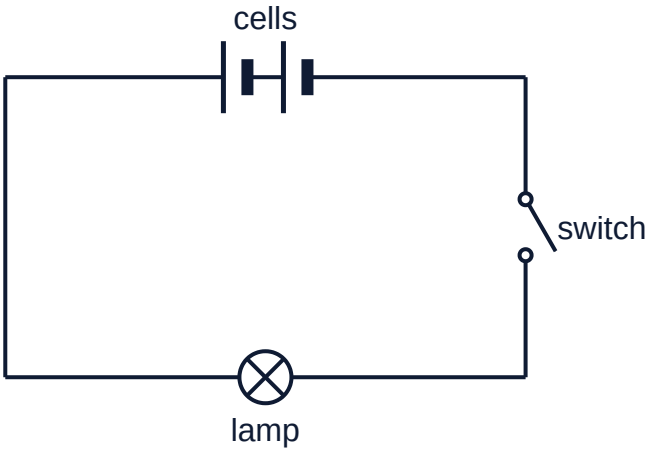
(a) Describe **two** ways the pupils can avoid being bitten by mosquitoes in the garden.

(b) The pupils pick tomatoes from the garden to eat. Explain why they should wash their hands and the tomatoes first. **[1]**

(c) Mariam scratches her arm on a rose thorn and it bleeds. Which of the body's defences against pathogens has been broken? **[1]**

- ☐ A) skin
- ☐ B) mucus
- ☐ C) stomach acid
- ☐ D) saliva

6 Youssef takes apart his toy torch. Inside there are two cells, a switch and a lamp. He draws this circuit diagram of the torch. **[1]**

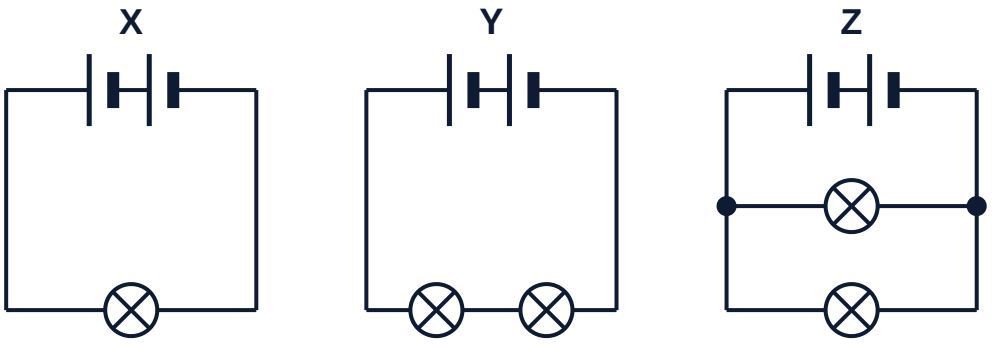


(a) Is the switch in Youssef's diagram open or closed?

(b) Explain why the lamp in Youssef's diagram is not lit. **[1]**

7 (continued) **[1]**

Youssef builds three circuits, X, Y and Z. All the cells and lamps are the same as in the torch.

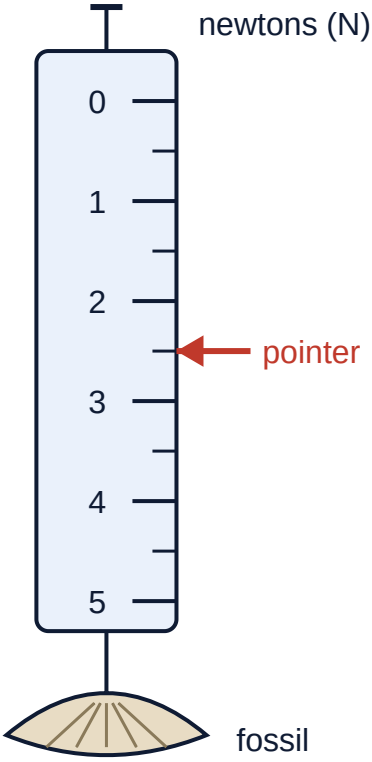


(c) In which circuit, X, Y or Z, are the lamps the dimmest?

(d) Youssef wants a torch with two lamps that are both as bright as the lamp in circuit X. Which circuit should he copy, Y or Z? **[1]**

(e) Explain why the lamps in circuit Z are brighter than the lamps in circuit Y. **[1]**

8 On a fossil hunt in the Western Desert, Hana finds a fossil sea urchin. At school she hangs it from a forcemeter. **[1]**



(a) What is the weight of the fossil? Read the forcemeter.

Working

Answer ____ N

(b) Gravity on Earth is 10 N/kg. Calculate the mass of the fossil in **grams**. **[1]**

Working

Answer ____ g

(c) A space robot takes the same fossil to the Moon, where gravity is weaker. Which statement is correct? **[1]**

- ☐ A) Its mass and its weight both stay the same.
- ☐ B) Its mass and its weight are both less.
- ☐ C) Its mass stays the same but its weight is less.
- ☐ D) Its mass is less but its weight stays the same.

(d) Gravity on the Moon is 1.6 N/kg. Calculate the weight of the fossil on the Moon. **[1]**

Working

Answer _____ N

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9 Hana's group also finds three rocks, J, K and L, in the Western Desert. **[1]**

Rock	What it looks like
J	small grains and tiny pieces of shell stuck together in layers; a shell fossil; crumbles when rubbed
K	interlocking crystals of different colours; no layers; very hard
L	crystals lined up in wavy bands; no fossils; very hard

(a) Which rock is igneous? Write J, K or L.

(b) What type of rock is L? **[1]**

(c) Rock J is a sedimentary rock. Describe how the shell fossil in rock J was formed. **[2]**

(d) Hana wants a quick test to help tell rock J from rock K. Which test is best? **[1]**

- ☐ A) Measure the temperature of each rock.
- ☐ B) Weigh each rock on a balance.
- ☐ C) Look at the colour of each rock.
- ☐ D) Drip water onto each rock and see if it soaks in.

10 At school, Hana drips vinegar onto a piece of rock J. The vinegar fizzes and bubbles of carbon dioxide gas are given off. **[1]**

Rock J + vinegar → new substances, including carbon dioxide

(a) What are the reactants in this reaction?

- ☐ A) carbon dioxide only
- ☐ B) rock J and vinegar
- ☐ C) bubbles and water
- ☐ D) rock J only

(b) Give the evidence that shows a chemical reaction happened. **[1]**

11 (continued) **[1]**

The science club mixes citric acid powder and bicarbonate of soda in a cup of water. They measure the temperature before and after.

Before: 21 °C	After: 13 °C

(c) By how many degrees did the temperature change?

Working

Answer ____ °C

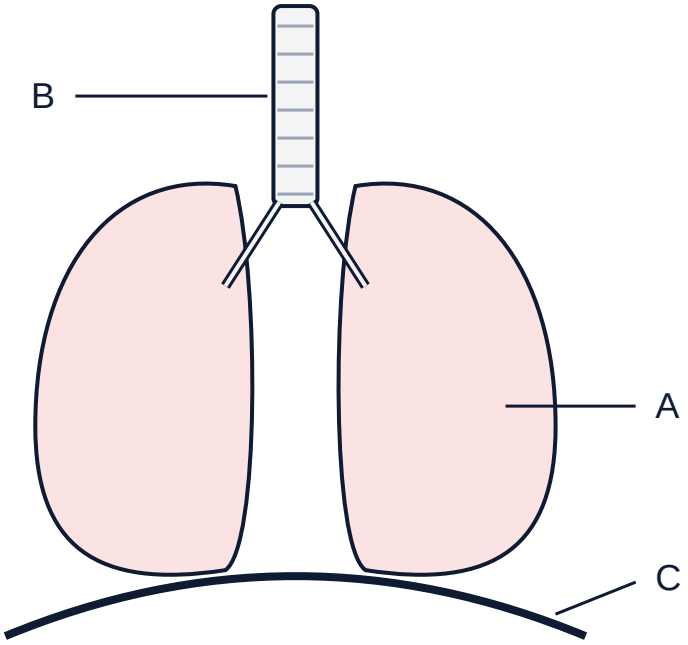
12 (continued) **[1]**

(d) Which observation is **not** evidence of a chemical reaction?

- ☐ A) The colour changes from blue to green.
- ☐ B) Bubbles of a gas are given off.
- ☐ C) The mixture gets hotter.
- ☐ D) Sugar disappears when it is stirred into water.

Paper 2 — 35 minutes — 40 marks

1 Omar's family visits the Red Sea coast. Omar goes snorkelling. He breathes air through a tube called a snorkel. The diagram shows part of his respiratory system. **[1]**



(a) Which letter labels the trachea (windpipe)?

(b) Name the gas that moves from the air in the lungs into the blood. **[1]**

2 (continued) **[1]**

Omar counts his breaths in one minute.

Activity	Breaths per minute
resting on the beach	16
after swimming hard	34

(c) By how much did his breathing rate increase?

Working

Answer _____ breaths per minute

(d) Explain why Omar breathes faster after swimming hard. **[1]**

3 (continued) **[1]**

Omar sees a dolphin come up to the surface of the sea to breathe. A dolphin is a mammal.

(e) Which statement is correct?

- ☐ A) Dolphins take oxygen from the water using gills, like fish.
- ☐ B) Dolphins have lungs and breathe air, like humans.
- ☐ C) Dolphins do not need oxygen because they live in water.
- ☐ D) Dolphins take in oxygen through their skin only.

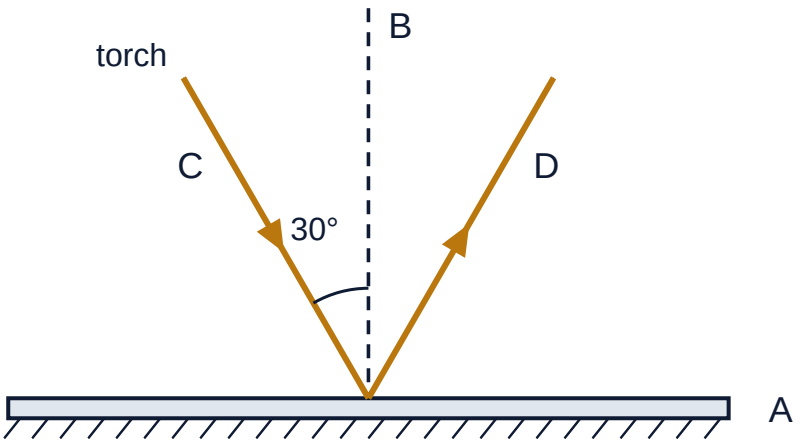
4 This question is about growing up. **[1]**

(a) Name the organs in the female reproductive system that make egg cells.

(b) Which statement about puberty is correct? **[1]**

- ☐ A) It starts at exactly the same age for everyone.
- ☐ B) It only happens to boys.
- ☐ C) It is the time when a child's body changes into an adult's body.
- ☐ D) It stops the body from growing taller.

5 On the boat, Omar shines a torch at a plane mirror. The diagram shows the ray of light. **[1]**



(a) Which letter labels the normal?

(b) What is the angle of reflection? **[1]**

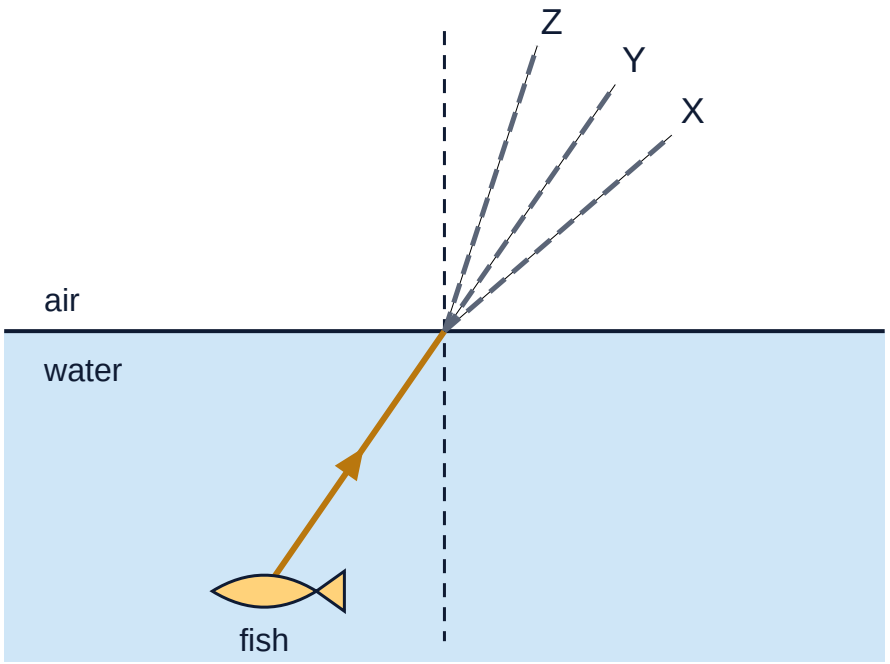
Working

Answer °

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6 (continued) **[1]**

Omar looks down at a fish from the side of the boat. Light from the fish travels up to the water surface. X, Y and Z show three possible paths for the light after it leaves the water.

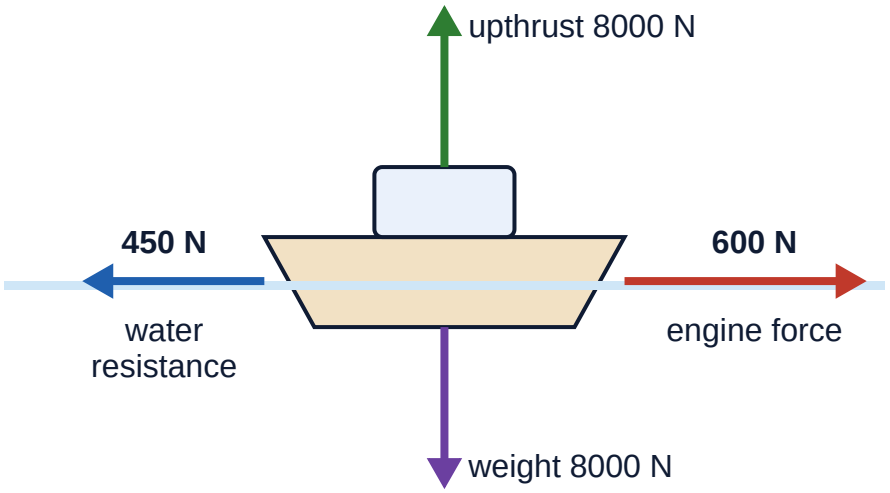


(c) What is the name for the change in direction of light when it passes from water into air?

(d) Which line, X, Y or Z, shows the path of the light in the air? **[1]**

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7 A glass-bottom boat moves across the sea. The diagram shows the forces on the boat. (The up and down arrows are not drawn to the same scale as the side arrows.) **[1]**



(a) Calculate the size of the overall (resultant) force on the boat in the forwards direction.

Working

Answer ____ N

(b) What happens to the speed of the boat? **[1]**

- ☐ A) It gets faster.
- ☐ B) It gets slower.
- ☐ C) It stays the same.
- ☐ D) The boat stops at once.

(c) Name the force that balances the weight of the boat. **[1]**

(d) The driver switches off the engine. Describe and explain what happens to the motion of the boat. **[1]**

8 On the beach, Laila puts a plastic bottle with its lid on into the sea water. She adds sand to the bottle, 100 g at a time, and measures how deep the bottle sits in the water. **[1]**

Sand added (g)	How deep the bottle sits (cm)	Floats or sinks?
0	2	floats
100	5	floats
200	8	floats
300	—	sinks

(a) How much deeper does the bottle sit with 200 g of sand than with no sand?

Working

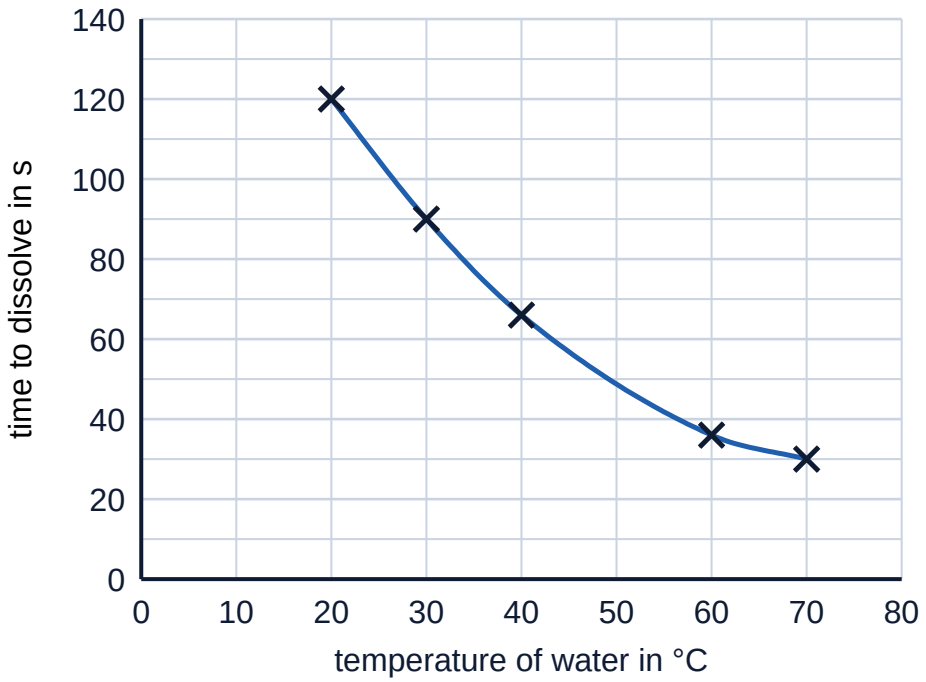
Answer ____ cm

(b) Why does the bottle sink when 300 g of sand is added? **[1]**

- ☐ A) The bottle changes shape.
- ☐ B) The water pushes up harder than before.
- ☐ C) The sand makes the water less salty.
- ☐ D) Its weight is now greater than the upthrust from the water.

(c) Laila wants to find out more exactly what mass of sand makes the bottle sink. Suggest how she should change her method. **[1]**

9 The science club investigates how the temperature of water affects the time a sugar cube takes to dissolve. Each time they use one sugar cube and 100 ml of water, and they stir at the same speed. They plot their results on a graph. **[1]**



(a) What type of scientific enquiry is this?

- ☐ A) observing over time
- ☐ B) fair testing
- ☐ C) identifying and classifying
- ☐ D) research using books

(b) What is the dependent variable in this investigation? (the variable they measure) **[1]**

(c) Use the graph. How long did the sugar cube take to dissolve in water at 30 °C? **[1]**

Working

Answer ____ s

(d) The club did not test water at 50 °C. Use the line on the graph to estimate how long the sugar cube would take to dissolve at 50 °C. **[1]**

Working

Answer ____ s

(e) Describe the pattern in the results. **[1]**

(f) Use the particle model to explain why sugar dissolves faster in hot water. **[2]**

10 Near the Red Sea coast there are salt pans. Sea water is let into wide, shallow pools. Over several weeks the water evaporates in the sunshine and salt is left behind. **[1]**

(a) Explain why getting salt back from sea water shows that dissolving is a reversible change.

(b) Which statement describes the evaporation in the salt pans? **[1]**

- ☐ A) It happens only at the surface of the water, below the boiling point.
- ☐ B) It happens only when the water reaches 100 °C.
- ☐ C) It makes large bubbles of gas all through the water.
- ☐ D) It changes the water into salt.

11 (continued) **[1]**

Omar heats 200 g of sea water in a dish until all the water has evaporated. 8 g of salt is left in the dish.

(c) What mass of water evaporated?

Working

Answer ____ g

12 (continued) **[1]**

The table shows the melting points and boiling points of three substances.

Substance	Melting point (°C)	Boiling point (°C)
mercury	−39	357
water	0	100
salt	801	1465

(d) Which substance is a solid at 500 °C?

(e) Omar says: "We can use a boiling point to help identify a pure substance." Why is he right? **[1]**

- ☐ A) All liquids boil at 100 °C.
- ☐ B) The boiling point depends on how much liquid there is.
- ☐ C) Each pure substance has its own boiling point.
- ☐ D) The boiling point changes every day.

13 Omar's uncle is a diver. He weighs his air tank on a balance. **[1]**

Empty tank: 12.5 kg	Tank filled with air: 15.3 kg

(a) What is the mass of the air in the full tank?

Working

Answer ____ kg

(b) What does this show about gases? **[1]**

14 (continued) **[1]**

The family keeps their drinks cold in a cool box on the beach.

(c) Which material is best for the walls of the cool box?

- ☐ A) copper
- ☐ B) steel
- ☐ C) plastic foam
- ☐ D) aluminium

15 The mountains near the Red Sea coast contain granite. Granite formed deep underground when magma cooled slowly. Rain, wind and waves break the granite into small grains, which are carried away to the beach and the sea. **[1]**

(a) What type of rock is granite?

(b) What is the name of the process in which the granite is worn away and the grains are carried to the sea? **[1]**

- ☐ A) melting
- ☐ B) erosion
- ☐ C) metamorphism
- ☐ D) solidification

(c) The grains settle on the sea floor. Describe how they could become sedimentary rock, and then later become metamorphic rock. **[2]**

16 At night on the beach, Omar sees a full moon. **[1]**

(a) Omar sees the full moon on 1 June. About when will he next see a full moon?

- ☐ A) 8 June
- ☐ B) 15 June
- ☐ C) 1 August
- ☐ D) 1 July

(b) Omar's sister says: "The Moon changes shape during the month because the Earth's shadow covers part of it." Explain the real reason the Moon's shape seems to change. **[1]**

(c) Which list shows these planets in order of distance from the Sun, starting with the closest? **[1]**

- ☐ A) Venus, Earth, Mars, Jupiter
- ☐ B) Earth, Venus, Mars, Jupiter
- ☐ C) Mars, Venus, Earth, Jupiter
- ☐ D) Venus, Mars, Earth, Jupiter